Homework 02

INT3405

October 3, 2024

Collaboration: List the names of all people you have collaborated with and for which question(s). You may discuss problems with other students, but you must not view or copy another student's written answers. Each student is responsible for independently understanding, writing, and submitting their own work.

Acknowledgments: If you reference solutions from published sources, the web, or other textbooks, you must acknowledge these sources. It's acceptable to use existing proofs for guidance, but this must be noted, and you should first attempt to solve the problem on your own. All students are expected to fully comprehend the steps they include in their written work.

Exercise 1. Given the following set of data (x, y):

$$\{(-1, 0), (0, 2), (1, 4), (2, 5)\}$$

a) Find the least square regression line $h(x) = w_0 + w_1 x$ by using Analytical solution.

b) Find the least square regression line $h(x) = w_0 + w_1 x$ by using the Gradient Descent (with 2 epochs, learning rate $\alpha = 0.01$, batch size $m = 4$, initial parameters $= 0$).

Exercise 2 [Ridge regression]. Write the Gradient Descent approach and Analytical solution for the Ridge Regression in details (using L2 in the $J(w)$, where $J(w)$ is MSE). Explain what happen when $\lambda \longrightarrow \infty$.

Exercise 3 [Multi-output linear regression]. Consider a linear regression model with a 2 dimensional response vector $y_i \in \mathbf{R}^2$. Suppose we have some binary input data $x_i \in \{0, 1\}$. The training data is as Table 1:

x	y
0	$(-1, -1)^T$
0	$(-1, -2)^T$
0	$(-2, -1)^T$
1	$(1, 1)^T$
1	$(1, 2)^T$
1	$(2, 1)^T$

Table 1: Data set

Let us embed each x_i into 2d using the following basis function:

$$\phi(0) = (1, 0)^T, \ \phi(1) = (0, 1)^T$$

The model becomes $h(x) = W^T\phi(x)$, where $W \in \mathbf{R}^{2\times 2}$ is a parameter matrix. Assume we use the MSE (mean squared error) as our cost function, find W that minimize the cost function.